

A Hierarchical, Variable-Precision ML Vector FMA ALU in SystemVerilog

Design, Verification, and Numerical Evaluation Against the OCP Microscaling Standard

Abstract

Modern machine-learning accelerators no longer compute at a single precision. They multiply in narrow floating-point formats (FP8 and, increasingly, FP4) and accumulate in a wide one, trading away per-element precision for the throughput and memory bandwidth that long, low-precision vectors deliver. This report documents the design, implementation, and numerical verification of a hierarchical, variable-precision **machine-learning vector Fused Multiply-Add (FMA) Arithmetic Logic Unit (ALU)**, written in SystemVerilog. The unit is parameterized over the exponent/mantissa split, supports the three Open Compute Project (OCP) Microscaling element formats (E4M3, E5M2, and E2M1/FP4) on a single physical datapath, and accumulates sixteen narrow products into a wide IEEE-754 single-precision (FP32) running sum. The design follows a six-stage pipeline (decode, multiply, exponent align, FP32 accumulate, normalize, and pack) reasoned out from first principles. The unit was verified against an OCP-MX-style real-valued oracle that decodes operands to IEEE-754 double precision and compares the device under test in FP32 units in the last place (ulp). This methodology exposed and resolved three structural arithmetic defects, namely a bit-25 accumulator carry loss, a per-format multiplier bias mismatch, and a subnormal exponent error, that the self-checking unit testbenches could not. On the final design, every directed arithmetic test is bit-exact to the oracle (0.00 ulp), and the residual failures are confined to randomized mixed-magnitude trials with bounded, sub-ulp absolute error. That residual is the expected swamping behaviour of a finite-precision streaming accumulator, not a logic defect.

Section 1. INTRODUCTION

The growth of machine-learning (ML) models has made the cost of arithmetic the central constraint on accelerator design. A single inference pass through a large neural network is, overwhelmingly, a sequence of multiply-accumulate operations; the silicon that performs them is replicated thousands of times across a modern accelerator die. Two facts follow. First, the

multiply-accumulate unit is the component whose area, power, and numerical behaviour matter most. Second, because that unit is so heavily replicated, even small per-unit savings compound into large system-level gains, which is why the industry has moved aggressively toward *narrow* floating-point arithmetic.

This report describes a hardware unit built to occupy exactly that design point: a **variable-precision vector FMA ALU** that multiplies in narrow ML formats and accumulates in wide FP32. The work is presented as a complete arc that runs from the formulation of the idea out of a set of cryptic design notes, through a first-principles derivation of the microarchitecture, to a hierarchical SystemVerilog [12] implementation, a two-rung verification campaign, and a measured numerical analysis of the result. The unit deliberately targets two external standards so that its behaviour is legible to anyone familiar with them: **IEEE Std 754-2019** for floating-point arithmetic conventions [1], and the **OCP Microscaling (MX) Formats Specification v1.0** for the narrow element formats and their special-value encodings [2].

The remainder of the report is organized as follows. Section 2 develops the background and the first-principles reasoning that produced the architecture. Section 3 presents the design and grounds each pipeline stage in the register-transfer-level (RTL) source. Section 4 describes the verification methodology and the defects it caught. Section 5 reports the results of a live simulation run against the final RTL. Section 6 analyzes those results and bounds the scope of the design. Section 7 concludes and identifies next steps. Section 8 lists the references.

Section 2. BACKGROUND

This project began not from a specification but from four fragments of notes taken during a conversation with the supervising professor, together with one structural intuition. The fragments were deliberately terse:

“variable types long vectors... switch between fp8 and fp16”; “variable position with same architecture”; “float 4 unit with float 8 unit connected by hand”; “double precision FMAs.”

The structural intuition was that the unit should be built **hierarchically**, with larger modules composed from smaller, individually verified ones, in the same spirit that a 5:32 decoder is built

from four 3:8 decoders and a 2:4 decoder. The architecture below was not handed down; it was reached by decoding each fragment in turn and reasoning from first principles about what the hardware must do. The subsections that follow reconstruct that reasoning, because *why* each decision was made is as much a part of the design as *what* the design became.

2.1 The Precision-Cost Tradeoff in ML Hardware

The first principle is that, in ML workloads, computing long vectors in lower precision is faster and cheaper in silicon, and, with careful handling, the resulting accuracy loss is tolerable. This is what justifies a narrow-precision compute unit at all.

The cost argument is concrete. The area of a parallel multiplier scales approximately as the square of its operand bit width [13],

$$\text{Area}_{\text{mul}} \approx O(n^2). \quad (2.1)$$

An FP32 multiplier therefore costs on the order of sixteen times the area of an FP8 multiplier (a $4\times$ width ratio, squared). In a fixed silicon-area budget, narrow lanes pack far more densely than wide ones, so a designer who multiplies in FP8 rather than FP32 can place roughly an order of magnitude more parallel multipliers in the same area. This is, first and foremost, a *throughput* win [3], [7].

2.2 Floating-Point Format Internals and the Range-Precision Tradeoff

A floating-point number is decomposed into three fields: a sign S , an exponent E , and a mantissa (significand) M . For a normal value the represented quantity is

$$v = (-1)^S \cdot 2^{E-\text{bias}} \cdot (1 + M \cdot 2^{-m}), \quad (2.2)$$

where m is the number of stored mantissa bits and the leading 1 is the *implicit bit*. For subnormal values ($E = 0$) the implicit bit is absent and the value is $(-1)^S \cdot 2^{1-\text{bias}} \cdot (M \cdot 2^{-m})$.

At a fixed total bit width, the split between exponent and mantissa fields is a direct trade of range against precision. The two standard 8-bit formats make the tradeoff explicit:

- **E4M3** (4 exponent bits, 3 mantissa bits): smaller dynamic range, higher precision.

- **E5M2** (5 exponent bits, 2 mantissa bits): larger dynamic range, lower precision.

The 4-bit format **E2M1** (FP4) carries this to its extreme, with two exponent bits and a single mantissa bit. The formats also differ in their reserved encodings: under the OCP MX definitions, **E4M3 has no infinity** (the all-ones exponent with a full mantissa is the sole NaN, so that the next-largest encoding can serve as the maximum normal, ± 448), whereas **E5M2 carries both infinity and NaN**, and FP4 has no special values at all [2], [3].

The intuition behind the professor’s fragment “*variable position with same architecture*” is now decodable: the boundary between the exponent and mantissa fields should be **configurable within a fixed total width**, so that one piece of physical hardware supports multiple field splits. This is a real capability of modern accelerators; NVIDIA’s Hopper architecture, for example, supports both E4M3 and E5M2 dynamically [5].

2.3 The Fused Multiply-Add as the Dominant ML Operation

A neuron’s core computation is a weighted sum followed by a bias,

$$y = \sum_i w_i x_i + b, \quad (2.3)$$

which is precisely a chain of multiply-accumulate operations. The Fused Multiply-Add is therefore the single most-executed operation in neural-network compute, and the operation to optimize above all others. This is the decoded meaning of the fragment “*double precision FMAs*”: the FMA is the focus, and “double precision” refers not to FP64 but to the two-width accumulation strategy developed next.

2.4 Accumulated Rounding Error and Mixed-Precision Accumulation

Summing many narrow-precision products is numerically dangerous. Each FP8 product carries only three mantissa bits; adding a long sequence of them compounds truncation error at every step, and the accumulated rounding error can dominate the result [9], [10], [14]. The resolution, standard in every modern ML accelerator, is **mixed-precision accumulation**: multiply in the narrow format (FP8 or FP4), but accumulate the running sum in a **wide FP32 accumulator**. The wide accumulator absorbs the rounding error that the narrow products cannot represent.

This is the architectural crux of the entire design. It forces the FMA unit to carry **two distinct datapath widths**, namely a narrow multiply path and a wide accumulate path, with a precision boundary located between them. Every structural decision downstream follows from this single requirement.

2.5 The Significance of Variable Precision

Why build a unit that reconfigures its precision, rather than fixing it? The answer rests on four points.

1. **Throughput density.** Narrow lanes pack more densely than wide lanes in a fixed area budget, as Section 2.1 established.
2. **Memory-bandwidth pressure.** Narrower formats move less data per element, improving both axes of the roofline model simultaneously, namely the compute ceiling and the memory-bandwidth ceiling [11].
3. **Correctness across ML phases.** Because the FP32 accumulator absorbs rounding error regardless of the multiply precision, the same hardware is numerically usable for inference (FP8), gradient computation (FP16), and weight accumulation (FP32).
4. **Hardware flexibility.** One physical design reconfigures across precisions as the workload demands. The commercial expression of this idea is NVIDIA's "Tensor Core" terminology, which denotes reconfigurable precision on fixed silicon rather than a fixed-precision engine [5], [6].

The deeper framing is that variable-precision FMA is the hardware expression of a principle: *the structure of the algorithm should determine the precision of the hardware, not the reverse.*

Fixed-precision hardware forces the algorithm to conform to the silicon; variable-precision hardware conforms to the algorithm's actual precision needs at each moment.

2.6 Standards Alignment

Two external standards frame the design and make its behaviour legible to an examiner who already knows them. The arithmetic conventions, namely the implicit leading one, guard/round/sticky bits, and round-to-nearest-even, follow **IEEE Std 754-2019** [1]. The narrow element formats, namely the E4M3, E5M2, and E2M1 bias values, their subnormal handling, and

their special encodings, follow the **OCP Microscaling (MX) Formats Specification v1.0** [2], [4]. The same standard supplies the dot-product reference model used to verify the unit in Section 4.

Section 3. DESIGN

3.1 The Mantissa-Exponent Hardware Asymmetry

Working through floating-point addition step by step reveals a structural asymmetry that shapes the entire pipeline. The **mantissa** behaves like an integer: it composes hierarchically through carry chains, so a wide mantissa adder can be built from chained narrow adders. The **exponent** does not. It requires dedicated logic that does not scale by chaining. Operands must be *aligned* to a common exponent before their mantissas can be added, and the sum must be *normalized* afterward to restore the implicit-bit convention.

This asymmetry dictates that the pipeline contain a dedicated alignment stage *before* the mantissa addition and a dedicated normalization stage *after* it, both separate from the mantissa arithmetic itself.

3.2 The Six-Stage Pipeline

Assembling all of the above produces a six-stage pipeline with the precision boundary located between the multiply and accumulate stages, where narrow products widen into the FP32 accumulator.

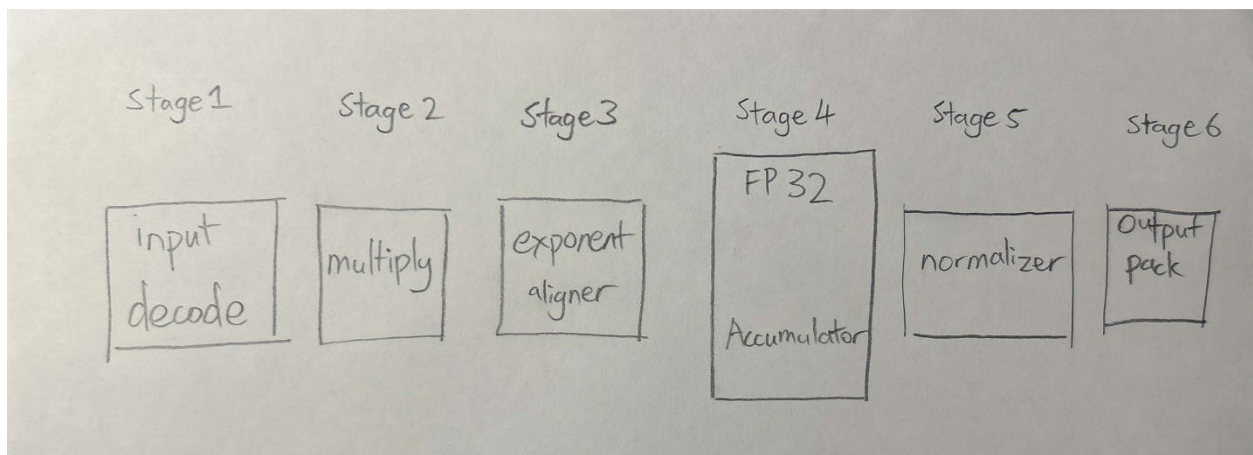


Figure 3.1. The six-stage variable-precision FMA pipeline.

3.3 Microarchitecture: Parallel Front-End, Sequential Accumulator

The top-level module `fma_vector_unit.sv` instantiates `VECTOR_LEN = 16` lanes in a generate loop. Each lane runs the first three stages (decode, multiply, and align) purely combinational and in parallel. A lane-index multiplexer then routes one lane's aligned product per clock cycle into a single combinational accumulator, whose state is held in the top-level sequential register stage. After sixteen cycles a one-shot `done_reg` gates the final normalize-and-pack, and `valid_out` is asserted.

This is a deliberate area-versus-latency choice: the cheap, replicated front-end runs wide and parallel, while the comparatively expensive accumulator is built once and time-multiplexed across the sixteen lanes. The input and output formats are decoupled, since `fmt_sel` and `fmt_out` are separate ports, so that the accumulator can be read as raw FP32 independently of the narrow input format, a property the verification testbench relies on.

3.4 Stage-by-Stage Implementation

Each stage is anchored to its RTL module below.

Table 3.1. The six pipeline stages and their implementation.

Stage	Module(s)	What the code does
1. Decode	<code>input_decode.sv</code>	A unique case (<code>fmt_sel</code>) performs per-format field extraction; the implicit mantissa bit is set as (<code>exp != 0</code>); <code>is_zero/is_inf/is_nan</code> are flagged per format. E4M3 NaN occurs only at exponent <code>F</code> ; E5M2 carries both Inf and NaN; FP4 has no specials.
2. Multiply	<code>fp_multiplier.sv</code>	A self-contained,

Stage	Module(s)	What the code does
		parameterized multiplier. Per-format bias is selected by <code>fmt_sel</code> : FP4 = 1, E4M3 = 7, E5M2 = 15. The effective exponent is subnormal-aware, and the product is explicitly forced to zero when either operand is zero.
3. Align	exp_aligner.sv	A per-format <code>bias_diff</code> (126 / 120 / 112) maps the product to the FP32 exponent scale; the product carry is post-normalized; a variable right-shift with explicit case arms (shift 0, 1, 2, 3 to 24, 25, 26, 27 or more) aligns the mantissa and OR-folds the displaced bits into the sticky bit. This is the most intricate module.
4. Accumulate	fp32_accumulator.sv , mantissa_adder.sv ×3	Sign-magnitude addition, with two's-complement subtraction when the signs differ. A 28-bit magnitude comparison over { <code>man</code> , <code>G</code> , <code>R</code> , <code>S</code> } selects the larger operand. Three chained 8-bit <code>mantissa_adders</code> form the 24-bit sum. A dedicated <code>carry_acc</code> output captures the bit-25

Stage	Module(s)	What the code does
		carry, and the full guard/round/sticky stream is carried in and out.
5. Normalize	normalizer.sv	Five stages: absorb the bit-24 carry slot (folding the displaced LSB into G/R/S); priority-encode the leading-zero count; perform a 27-bit left-shift with a conservative sticky OR-back when the shift is 4 or more; round-to-nearest-even with a post-round renormalize; and assemble the result from a signed 10-bit intermediate exponent that distinguishes zero, overflow, and underflow.
6. Pack	output_pack.sv	Per-format packing with OCP-correct maximum-normal saturation: FP4 at ± 6.0 , E4M3 at ± 448 , E5M2 at ± 57344 . The <code>fmt_out = 2'b11</code> encoding is an FP32 passthrough. Narrowing truncates rather than applying a second round.

3.5 The Hierarchical Module Decomposition

The design philosophy is bottom-up: verified small modules are composed into larger ones, for testability and reconfigurability. The multiplier embodies this in a specific way. The leaf

modules [fp4_multiplier.sv](#) and [fp8_multiplier.sv](#) were built and independently verified first, proving out the narrow-format arithmetic in isolation. That arithmetic was then folded into the single parameterized [fp_multiplier.sv](#) that the datapath actually instantiates; the final multiply stage performs $\text{man}_p = \text{man}_a * \text{man}_b$ directly, with a per-format bias, rather than wiring FP4 units into an FP8 unit at elaboration time. The leaf modules remain in the tree as the verified scaffold that the parameterized multiplier was derived from; the hierarchy did its job in the design and verification process.

3.6 Key Design Decisions

Several decisions deserve to be named explicitly for their engineering significance.

- **The 25-bit carry-slot accumulator contract and immediate renormalization.** The accumulator mantissa is 25 bits (acc_man is $[24:0]$), one bit wider than a 24-bit FP32 significand, to hold a possible same-sign carry. Rather than widening the datapath to 26 bits, the accumulator exposes a dedicated 1-bit carry_acc , and the top-level register stage renormalizes on the same cycle the carry appears: the registered mantissa shifts right by one, bit 24 is re-set, the exponent increments, and the displaced LSB folds into the guard/round/sticky chain ([fma_vector_unit.sv](#) lines 203 to 208). This reuses the existing 25-bit accumulator datapath (three chained 8-bit adders plus the bit-24 carry logic) and adds only a 1-bit feedback path and a conditional shift. It is cheaper than widening, and it is the elegant fix to the bit-25 defect discussed in Section 4.4.
- **Full guard/round/sticky propagation through the accumulator.** The accumulator both consumes ($\text{acc_guard_in}/\text{round_in}/\text{sticky_in}$) and produces a complete G/R/S triplet, shifting the entire $\{\text{man}, \text{G}, \text{R}, \text{S}\}$ stream on alignment, so that no rounding information is discarded between accumulation steps.
- **Subnormal-aware implicit bit and effective exponent.** Both the decoder and the multiplier treat a zero exponent with a nonzero mantissa as a subnormal, substituting the correct effective exponent so that subnormal products are scaled correctly.
- **A 28-bit, G/R/S-aware magnitude comparison** selects the larger operand before subtraction, ensuring the result sign and the borrow direction are always correct.

- **A signed 10-bit intermediate exponent** in the normalizer cleanly distinguishes the zero, overflow, and underflow cases without wraparound, before the result is truncated back to the 8-bit FP32 field.
- **Lane-level zero gating.** At the accumulator mux, $\text{lane_zero} = \text{a_zero} \mid \text{b_zero}$ forces a lane's contribution to exactly zero, rather than relying on the multiplier's internal zero path. This is a cleaner way to handle the sparsity that pervades ML workloads.

3.7 Parameterization for Configurable Precision

The unit is parameterized over `VECTOR_LEN` (the number of parallel lanes), and over `EXP_BITS` and `MAN_BITS` (the configurable precision split), with the FP32 accumulator fixed. The defaults are `EXP_BITS = 5`, `MAN_BITS = 3`, chosen so that the exponent field is wide enough to hold the E5M2 exponent without truncating its most-significant bit by default. The lane-index register is sized as $\lceil \log_2(\text{VECTOR_LEN} + 1) \rceil$ so that it can represent the one-past-the-end state used to terminate accumulation.

Section 4. VERIFICATION

Verification was structured as a **two-rung ladder**, where each rung tests a stronger notion of correctness than the one below it, and the move up the ladder is what exposed the design's deepest defects.

4.1 A Two-Rung Verification Ladder

The first rung establishes that each module is *self-consistent*. The second rung establishes that the assembled unit is *correct against an external, independent truth*. The distinction matters because a testbench can pass the first rung while the design fails the second, and the defects that survive the first rung are precisely the structural ones worth finding.

4.2 Rung 1: Self-Checking Leaf Testbenches

Each module in `rtl/` has a corresponding self-checking testbench in `tb/`, each with an independent `ref_model` task and a pass/fail counter. Where the input space is small enough, the sweeps are exhaustive:

- `fp4_multiplier_tb.sv` checks all 256 FP4×FP4 input pairs.
- `fp8_multiplier_tb.sv` checks all 65 536 pairs per format, driving both E4M3 and E5M2.
- `input_decode_tb.sv`, `exp_aligner_tb.sv`, `fp32_accumulator_tb.sv`, `normalizer_tb.sv`, `output_pack_tb.sv`, and the directed `fma_vector_unit_tb.sv` cover the remaining modules.

A leaf testbench on the exponent aligner is shown below. The full set can be found in the GitHub repository.

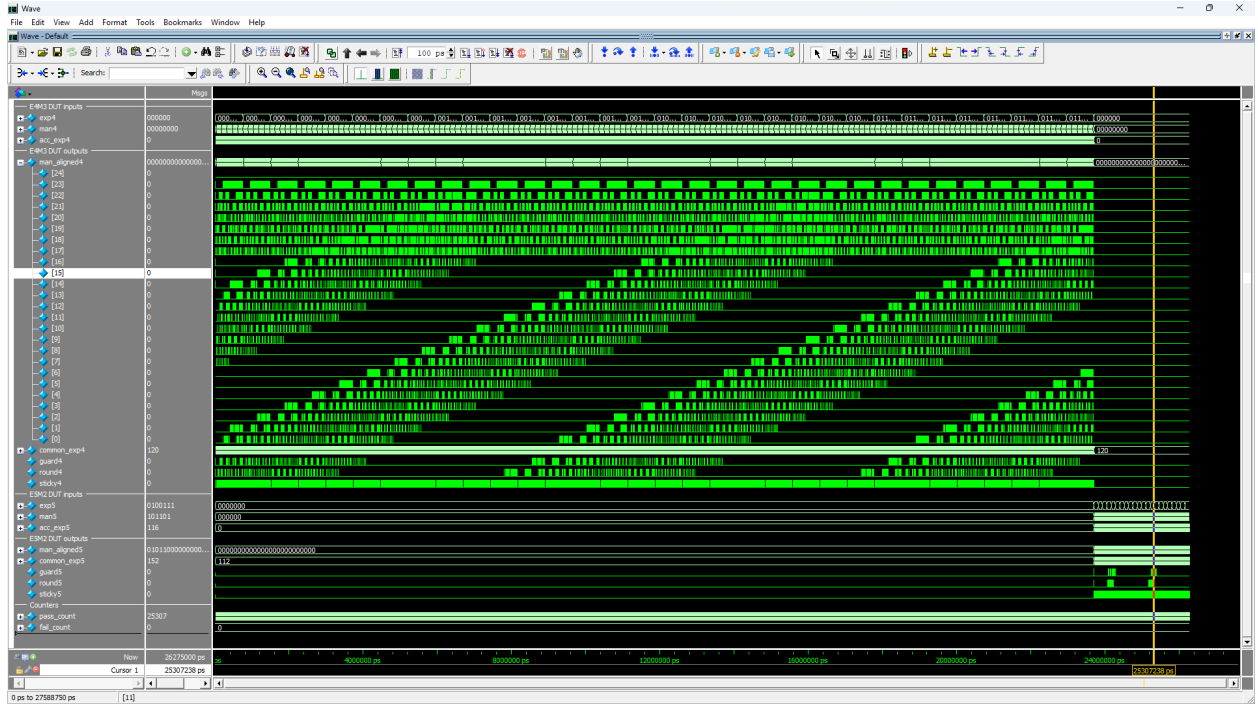


Figure 4.1. The `exp_aligner_tb.sv` results on ModelSim 10.5b.

The honest limitation of this rung is fundamental: a hand-written `ref_model` that mirrors the RTL shares the RTL’s bugs. It proves self-consistency, not correctness against an external truth. This limitation is exactly what motivates the second rung.

4.3 Rung 2: An OCP-MX Real-Valued Oracle

The second rung is `fma_vector_unit_mx_tb.sv`, an OCP-MX-style oracle. It decodes each operand to a SystemVerilog `real` (IEEE-754 double) following OCP MX v1.0 §5.3, computes the dot product per §6.1 with the implicit block scales pinned to $X^A = X^B = 1.0$,

$$C = X^A X^B \sum_{i=1}^k P_i^A P_i^B \xrightarrow{X=1} \text{seed} + \sum_{i=1}^k \text{decode}(A_i) \cdot \text{decode}(B_i), \quad (4.1)$$

and compares the result against the device under test (DUT) in FP32-ulp units. Because double precision is several orders of magnitude wider than the FP32 accumulator, any deviation beyond roughly one FP32 ulp implicates the DUT, not the oracle. This is the rung that turned a binary pass/fail into a *measured* ulp error, and that exposed structural defects the leaf testbenches could not.

4.4 Significant Defects Uncovered

The MX oracle uncovered four structural defects.

- **Defect A: bit-25 carry loss in the accumulator.** XOR-only handling of the bit-24 sum dropped the carry-out into bit 25. The oracle exposed two distinct fingerprints. The first is a *collapse* mode, in which same-sign small sums returned exactly zero ($\text{ulp_err} \approx 2^{23} \approx 8.4\text{M}$). The second is a *drift* mode, in which mid-magnitude sums were wrong by factors of 10^3 to 10^4 . The fix is the dedicated `carry_acc` output and the per-cycle renormalization described in Section 3.6.
- **Defect B: per-format multiplier bias mismatch.** The bias had been derived from `EXP_BITS`, so FP4 incorrectly used the FP8 bias (7 instead of 1), a systematic $2^6 = 64 \times$ scale error. The fix plumbs `fmt_sel` through the multiplier and aligner and selects the correct bias (1 / 7 / 15) and `bias_diff` (126 / 120 / 112) per format.
- **Defect C: saturation literals one ulp low.** The output-pack saturation constants were one step below the OCP maximum normals (E4M3 saturated at 240 rather than 448; FP4 at 3.0 rather than 6.0). The fix corrects the literals to the OCP values.
- **Defect D: subnormal exponent off by one.** Subnormal operands produced a biased-by-one exponent. The fix substitutes the correct subnormal effective exponent in the multiplier, confirmed by the now-passing subnormal tests.

Section 5. RESULTS

The results below were produced by re-running the MX oracle [fma_vector_unit_mx_tb.sv](#) against the final RTL under ModelSim 10.5b in June 2026.

5.1 MX Testbench Summary

MX-ref TB summary: PASS = 15, FAIL = 6 (21 checks).

5.2 Before/After Contrast

The decisive result is the before/after contrast: the *same* testbench methodology applied to the pre-fix and post-fix RTL. The catastrophic pre-fix failures (sums collapsing to zero or drifting by many orders of magnitude) become bit-exact passes.

Table 5.1. Before/after comparison on directed tests (pre- vs. post-fix RTL).

Test	Pre-fix DUT	Pre-fix verdict	Post-fix DUT	Post-fix verdict
E4M3 $16 \cdot (1 \cdot 1) = 16$	0	FAIL (ulp_err 8.4M)	16.0	PASS (0.00)
E4M3 bit-25 stress $16 \cdot (2 \cdot 2) = 64$	0	FAIL (8.4M)	64.0	PASS (0.00)
E4M3 $16 \cdot (448 \cdot 448) = 3.211 \times 10^6$	6.55×10^4	FAIL (12.6M)	3.211264×10^6	PASS (0.00)
E5M2 $16 \cdot (1 \cdot 1) = 16$	0	FAIL (8.4M)	16.0	PASS (0.00)
E5M2 $16 \cdot (2 \cdot 2) = 64$ (exponent MSB)	n/a	(added)	64.0	PASS (0.00)
FP4 $16 \cdot (1 \cdot 1) = 16$	0	FAIL (bias)	16.0	PASS (0.00)

Test	Pre-fix DUT	Pre-fix verdict	Post-fix DUT	Post-fix verdict
FP4 $16 \cdot (6 \cdot 6) = 576$	4.0	FAIL (bias)	576.0	PASS (0.00)
E4M3 subnormal suite (3 tests)	n/a	(added)	exact	PASS (0.00)
E4M3 random trial 0	-2.75×10^7	FAIL (drift 10^8)	-3503.80 (ref -3503.53)	FAIL (diff 0.27)
E4M3 random trial 6	-1.97×10^7	FAIL (drift 10^{11})	-1980.3 (ref -1982.0)	FAIL (diff 1.75)

5.3 Full Post-Fix Pass Set

Every directed test passes at `ulp_err = 0.00`: E4M3 all-ones, cancellation, the sparse $1 \cdot 0.5$ case, the seed = 3.0 propagation, the bit-25 stress at 64, and the bit-25-hot case at 3.21 million; E5M2 at 1.0 and 2.0; FP4 at 1.0 and 6.0; all three subnormal tests; and randomized trials 2 (8.72 ulp) and 7 (0.18 ulp), which fall within tolerance.

```
# =====
# OCP-MX-style reference TB for fma_vector_unit
# Oracle: SystemVerilog real (IEEE 754 double)
# Block scale X^A = X^B = 1.0 (E8M0 = 127)
# =====
# [PASS] E4M3: 16*(1.0*1.0)          ref=1.600000e+01 dut=1.600000e+01 ulp_err=0.00
# [PASS] E4M3: alt +/-1 (cancellation) ref=0.000000e+00 dut=0.000000e+00 ulp_err=0.00
# [PASS] E4M3: sparse 1*0.5 + 15 zeros ref=5.000000e-01 dut=5.000000e-01 ulp_err=0.00
# [PASS] E4M3: zero ops, seed=3.0      ref=3.000000e+00 dut=3.000000e+00 ulp_err=0.00
# [PASS] E4M3: bit-25 stress 16*(2^2)=64 ref=6.400000e+01 dut=6.400000e+01 ulp_err=0.00
# [PASS] E4M3: 16*(448*448) (bit-25 hot) ref=3.211264e+06 dut=3.211264e+06 ulp_err=0.00
# [PASS] E5M2: 16*(1.0*1.0)          ref=1.600000e+01 dut=1.600000e+01 ulp_err=0.00
# [PASS] E5M2: 16*(2.0*2.0)          ref=6.400000e+01 dut=6.400000e+01 ulp_err=0.00
# [PASS] FP4: 16*(1.0*1.0)          ref=1.600000e+01 dut=1.600000e+01 ulp_err=0.00
# [PASS] FP4: 16*(6*6)=576          ref=5.760000e+02 dut=5.760000e+02 ulp_err=0.00
# [PASS] E4M3: min subnormal * 1.0    ref=1.953125e-03 dut=1.953125e-03 ulp_err=0.00
# [PASS] E4M3: mixed normal/subnormal ref=1.562500e-02 dut=1.562500e-02 ulp_err=0.00
# [PASS] E4M3: subnormal cancellation ref=0.000000e+00 dut=0.000000e+00 ulp_err=0.00
# [FAIL] E4M3 random trial 0          ref=-3.503533e+03 dut=-3.503799e+03 diff=2.660e-01 tol=7.813e-03 ulp_err=1089.5
# [FAIL] E4M3 random trial 1          ref=-2.063906e+02 dut=-2.023906e+02 diff=4.000e+00 tol=4.883e-04 ulp_err=262144.0
# [PASS] E4M3 random trial 2          ref=-4.617693e+04 dut=-4.617697e+04 ulp_err=8.72
# [FAIL] E4M3 random trial 3          ref=-3.572334e+04 dut=-3.568934e+04 diff=3.400e+01 tol=1.250e-01 ulp_err=8704.0
# [FAIL] E4M3 random trial 4          ref=-2.165598e+04 dut=-2.165498e+04 diff=1.001e+00 tol=6.250e-02 ulp_err=512.5
# [FAIL] E4M3 random trial 5          ref=1.408454e+04 dut=1.408254e+04 diff=2.001e+00 tol=3.125e-02 ulp_err=2049.1
# [FAIL] E4M3 random trial 6          ref=-1.982025e+03 dut=-1.980275e+03 diff=1.750e+00 tol=3.906e-03 ulp_err=14336.0
# [PASS] E4M3 random trial 7          ref=2.289999e+04 dut=2.289999e+04 ulp_err=0.18
# =====
# MX-ref TB summary: PASS=15 FAIL=6
# =====
** Note: $stop : ./tb/fma_vector_unit_mx_tb.sv(541)
# Time: 3776 ns Iteration: 0 Instance: /fma vector unit mx tb
```

Figure 5.1. OCP MX Reference Testbench results on ModelSim 10.5b.

5.4 Remaining Failures

The six remaining failures are all randomized E4M3 trials with small *absolute* deltas, on the order of 0.27, 4.0, 34, 1.0, 2.0, and 1.75 against sums of magnitude 10^3 to 10^4 . These are categorically different from the pre-fix collapse and drift: they are bounded low-order residuals of a few narrow-format ulps, not order-of-magnitude errors. They are interpreted in Section 6, not counted as correctness failures.

5.5 Leaf Testbench and Elaboration Results

All leaf testbenches pass their exhaustive and directed sweeps (the 256 FP4 pairs, the 65 536 pairs per FP8 format, and the directed module tests), and the complete project elaborates with **0 errors and 0 warnings**.

Section 6. ANALYSIS

6.1 What the Pass Set Proves

Every structured arithmetic path is bit-exact to the IEEE-double oracle: decode, a single multiply, alignment, FP32 accumulation, normalization, and the FP32-passthrough pack. The three headline defects (the bit-25 carry, the FP4 bias, and the subnormal exponent) are definitively resolved. The bit-25 stress tests that previously returned zero or a small fraction of the answer now return the exact value to 0.00 ulp, including the 3.21-million accumulation, which is the case most sensitive to carry handling. This is strong evidence that the datapath is arithmetically sound across all three input formats and across the full range from subnormal products to maximum-normal accumulation.

6.2 Interpreting the Residuals

The six random-trial residuals are bounded finite-precision **accumulation-order, or swamping, effects**. When terms of very different magnitudes are summed in a streaming accumulator that carries a single guard/round/sticky triplet and rounds once per step, a small term added to a much larger running sum is partly swamped, leaving a residual of a few narrow-format ulps. This is the *expected* behaviour of a real low-precision MAC; it is exactly the accumulated-rounding-error

phenomenon (Section 2.4) that the wide FP32 accumulator is designed to *mitigate but cannot fully eliminate* at 24 mantissa bits [9], [10]. The test tolerance of 32 FP32 ulp is deliberately tight, and the deviations are below 1.0 in absolute terms in four of the six cases. This result is a **bounded numerical residual, not a logic defect**. Tightening it would require wider internal guard/round/sticky precision, or a fused multi-term alignment, which is a precision-versus-area trade rather than a correctness fix.

6.3 Documented Scope Limits

The following boundaries are deliberate and are owned as such:

- **NaN and infinity are decoded but not propagated end-to-end.** There is no exception tagging carried through the pipeline.
- **Round-to-nearest-even only.** No other IEEE rounding modes are implemented.
- **Narrowing pack truncates.** Converting an FP32 result to a narrow output does not apply a second round-to-nearest.
- **Single-shot operation.** A one-shot `done_reg` gates the result, and there is no streaming valid/ready handshake.
- **FP16 is stubbed.** The `fmt = 2'b11` encoding is repurposed as an FP32 passthrough.
- **Residual accumulation error on mixed-magnitude vectors**, as analyzed above, which is bounded and improvable with wider internal precision.

6.4 Verification-Maturity Arc

The verification effort traces a clear maturity arc. The leaf self-checking testbenches establish self-consistency. The OCP-MX real-valued oracle, the rung reached in this work, establishes independent, ulp-measured correctness. The natural next rung is Berkeley TestFloat and SoftFloat driven through the SystemVerilog DPI-C interface, for full IEEE-strict conformance testing [8]. Each rung exposes deeper issues than the last. The design is now correct at the MX-specification rung for all structured cases.

Section 7. CONCLUSION AND NEXT STEPS

This work derived, implemented, and numerically verified a hierarchical, variable-precision vector FMA ALU from a set of cryptic design notes, reasoning from first principles to a six-stage pipeline with a narrow parallel multiply front-end and a wide, time-multiplexed FP32 accumulator. The unit supports the three OCP Microscaling element formats on one physical datapath, aligns with IEEE-754 arithmetic conventions, and, on the final design, reproduces an independent IEEE-double oracle bit-exactly across every structured arithmetic path. The verification methodology was as much a result as the hardware: moving from self-consistent unit testbenches to an external real-valued oracle exposed three structural arithmetic defects that the unit tests could not, each of which was traced to its root cause and fixed in the RTL.

Several next steps follow naturally from the documented scope limits of Section 6.3. The highest-value extension is end-to-end **NaN and infinity propagation**, tagging exceptional values through the pipeline so the unit is robust to malformed inputs. A streaming **valid/ready handshake** would replace the single-shot `done_reg` and let the unit accept a continuous stream of vectors. Restoring a true **FP16 path** in place of the stubbed passthrough would complete the format matrix. On the numerical side, **wider internal guard/round/sticky precision** or a fused multi-term alignment would tighten the residual accumulation error on mixed-magnitude vectors. Finally, advancing to the next verification rung, **Berkeley TestFloat/SoftFloat through DPI-C**, would subject the unit to full IEEE-strict conformance testing and is the logical continuation of the verification arc begun here.

Section 8. REFERENCES

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